

Anndeen Daley

August 13, 2026

CS 619 Data Mining

Project Proposal

Dataset:

The dataset I chose to analyze is ““Estimation of Obesity Levels Based on Eating Habits and Physical Condition,” located on the UCI Machine Learning Repository. It contains over 2,100 records describing the eating habits, physical activity, and demographics of individuals from Mexico, Peru, and Colombia.

<https://archive.ics.uci.edu/dataset/544/estimation+of+obesity+levels+based+on+eating+habits+and+physical+condition>

These are the key attributes:

Attribute	Type	Description
Gender	Categorical	Female / Male
Age	Numeric	Age
Height	Numeric	Height (meters)
Weight	Numeric	Weight (kilos)
family_history_with_overweight	Binary (y/n)	Has a family member suffered/suffers from overweight?
FAVC	Binary (y/n)	Do you eat high caloric food frequently?
FCVC	Numeric (1–3)	Do you usually eat vegetables in your meals?
NCP	Numeric	How many main meals do you have daily?
CAEC	Categorical	Do you eat any food between meals? (No / Sometimes / Frequently / Always)
SMOKE	Binary(y/n)	Do you smoke?
CH2O	Numeric	How much water do you drink daily?
SCC	Binary (y/n)	Do you monitor the calories you eat daily?
FAF	Numeric	How often do you have physical activity?
TUE	Numeric	How much time do you use technological devices such as cell phone, videogames, television, computer and others?
CALC	Categorical	How often do you drink alcohol?
MTRANS	Categorical	Which transportation do you usually use?
NObesidad (class) Target	Categorical (7 levels)	Obesity level

Problem:

The question I hope to answer during this project is can a person's obesity category be predicted from their eating habits and lifestyle rather than from direct BMI? And also analyze whether there are unclear associations with eating and activity habits that always occur with higher obesity categories like consumption of high-caloric food together with low physical activity is associated with a specific obesity level.

Approaches:

I will use classification learning as the main approach, since the target attribute (NObeyesdad) is a separate class with seven possible values and the goal is to predict it from other attributes. I plan to use two classifiers: J48 and Naïve Bayes/Random Forest. Also Apriori because it will look for combinations of lifestyle habits that tend to occur together often and reliably.

Transformations:

- Save the dataset as an ARFF file, then split it into training (80%) and test (20%) sets using Weka
- I will spot-check the synthetic (SMOTE generated) records for obvious duplicates etc.
- Exclude height and weight from the input attributes, they will be a secondary comparison.
-